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SELECT ACADEMIC PROJECTS

Texas Groundwater Quality Database & Geospatial Analysis (MySQL, Python)

- Designed and implemented a normalized MySQL relational database (64 tables, 3NF) for Texas Water Development Board groundwater-quality data across 254 counties.
- Cleaned and standardized fragmented datasets containing mineral concentrations (arsenic, fluoride, lithium, uranium) spanning two decades; resolved unit inconsistencies and geographic alignment issues.
- Developed complex SQL queries with multi-table joins for county-level aggregation and mineral co-occurrence analysis.
- Integrated database with Python (pandas, GeoPandas, Folium) to create choropleth maps identifying geographic patterns and contamination hotspots for public health research.

Unsupervised Learning: Clustering & Association Rule Mining (Python)

- Implemented and compared 6 clustering algorithms (K-means, Hierarchical Agglomerative Clustering, GMM, DBSCAN, Spectral, Graph-based) on 100K health/lifestyle records.
- Evaluated cluster quality using internal metrics (Silhouette, Davies-Bouldin, Calinski-Harabasz) and external metrics (ARI, NMI); visualized results with PCA projections.
- Conducted association rule mining on 541K retail transactions using Apriori and FP-Growth algorithms; discovered 79 actionable rules with lift up to 24 for product bundling strategies.
- Analyzed threshold sensitivity and identified high-confidence product trios for merchandising recommendations.

Frito-Lay Employee Attrition Analysis (R)

- Analyzed >1,000 HR records using R (tidyverse, caret) to identify drivers of turnover.
- Built Naive Bayes and KNN models achieving 78% accuracy; performed significance testing and cost modeling.

Predicting Crab Age (R)

- Modeled 25,000 crab measurements; engineered polynomial, interaction, and transformed features.
- Used stepwise and all-subsets regression to reduce multicollinearity, improving MAE to 0.15.
- Built R Shiny visual dashboards to display relationships between physiology and age.

Housing Price Prediction (R)

- Developed multivariate regression models with engineered features (log terms, interactions, normalization) achieving adjusted $R^2 = 0.87$.
- Cleaned and imputed 80+ variables; validated assumptions and generated Kaggle-ready predictions.

Heart Disease Modeling (R & Python)

- Analyzed >300,000 CDC BRFSS records to engineer health indices and identify predictors of heart disease.
- Compared models (KNN, LDA, QDA, logistic regression), achieving AUC > 0.80.
- Used ROC curves, confusion matrices, cross-validation, and feature importance for interpretability.

Hospital Length of Stay Prediction (R)

- Modeled patient LOS using linear regression, LASSO, kNN, and Random Forests.
- Identified operational and demographic predictors (infection risk, nursing ratios).

Diabetes EDA & PCA/LDA (Python)

- Visualized and analyzed Pima Indians dataset with pandas, scikit-learn, seaborn.
- Built PCA and LDA models and evaluated variance explained vs. class separation.

Classification Model Pipeline Optimization (Python)

- Built cross-validated pipelines for logistic regression, SVMs, tree-based models.
- Tuned hyperparameters using GridSearchCV and RandomizedSearchCV; assessed decision boundaries and calibration.